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SEMI E94.1-0312 (Reapproved 0324)
SPECIFICATION FOR SECS-II PROTOCOL FOR CONTROL JOB MANAGEMENT (CJM)

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1 Purpose

1.1 This Specification maps the services and data of SEMI E94 to SECS-II streams and functions and data definitions.

2 Scope

2.1 This is a specification covering equipment supporting automated control job management.

2.2 This Document applies to all implementations of SEMI E94 that use the SECS-II message protocol (SEMI E5). Compliance to this Standard requires compliance to both SEMI E94 and SEMI E5.

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3 Limitations

3.1 This Specification applies to semiconductor equipment that also uses SEMI E40 Process Jobs.

4 Referenced Standards and Documents

4.1 *SEMI Standards and Safety Guidelines*

SEMI E5 — Specification for SEMI Equipment Communications Standard 2 Message Content (SECS-II)

SEMI E30 — Specification for the Generic Model for Communications and Control of Manufacturing Equipment (GEM)

SEMI E39.1 — Specification for SECS-II Protocol for Object Services (OSS)

SEMI E40 — Specification for Processing Management

SEMI E94 — Specification for Control Job Management

NOTICE: Unless otherwise indicated, all documents cited shall be the latest published versions.

5 Terminology

5.1 None.

6 Mapping

6.1 This section shows the specific SECS-II streams and functions that shall be used for SECS-II implementation of the services defined in SEMI E94, as well as the parameter mapping for data attached to services.

6.2 *Message Mapping*

6.2.1 *Services Message Mapping*

6.2.1.1 Table 1 defines the relationships between SEMI E94 services and SECS-II messages.

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Table 1 Services Message Mapping Table

Service Name	*Stream, Function*	*SECS-II Message Name*
Create Object	S14,F9/10	Create Object Request/Acknowledge (SEMI E39.1)
CJAbort	S16,F27/28	Control Job Command Request/Acknowledge
CJCancel	S16,F27/28	Control Job Command Request/Acknowledge
CJDeselect	S16,F27/28	Control Job Command Request/Acknowledge
CJHOQ	S16,F27/28	Control Job Command Request/Acknowledge
CJPause	S16,F27/28	Control Job Command Request/Acknowledge
CJResume	S16,F27/28	Control Job Command Request/Acknowledge
CJStart	S16,F27/28	Control Job Command Request/Acknowledge
CJStop	S16,F27/28	Control Job Command Request/Acknowledge

6.2.2 *Event Message Mapping*

6.2.2.1 Table 2 defines the relationships between SEMI E94 collection events and SECS-II messages.

Table 2 Event Message Mapping Table

Event Name	*Stream, Function*	*SECS-II Message Name*
All state model transitions	If SEMI E30 style events: S6,F11/12	If SEMI E30 style events: Event Report Send/Acknowledge

6.2.3 *Parameter Mapping*

6.2.3.1 Table 3 defines the relationships between SEMI E94 service parameters and SECS-II data definitions.

Table 3 Parameter to SECS-II Data Items Mapping

Parameter Name	*Range*	*SECS-II Data Item*
ACKcode	TRUE, FALSE	ACKA
Action	SAVEJOBS, REMOVEJOBS	CPVAL (U1) 0 = SAVEJOBS. This command does not destroy the Process Jobs specified by this Control Job. 1 = REMOVEJOBS. This command destroys all Process Jobs specified by this Control Job.
CtrlJobID	1–80 characters (Conforms to ObjID in SEMI E39.1, § 6.)	OBJID
ErrorCode	Enumerated	ERRCODE
ErrorInfo	Error	L,2 1. ErrorCode 2. ErrorText
ErrorText	1–80 characters	ERRTEXT
Status	Acknowledgement and error	L,2 1. ACKcode 2. ErrorInfo

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6.2.4 *SECS-II Data Items without Corresponding SEMI E94 Parameters*

6.2.4.1 Table 4 contains the SECS-II data items that do not correspond to SEMI E94's service parameter.

Table 4 Additional Data Item Requirements Table

Function	*SECS-II Data Item*
Used to identify control job commands.	CTLJOBCMD
Used to satisfy SECS-II conventions for linking a multi block inquiry with subsequent multi block message.	DATAID

6.2.5 *Variable Data Item Mapping*

6.2.5.1 Table 5 shows the specific SECS-II data classes, and formats needed for SECS-II implementations of SEMI E94 variable data items.

Table 5 Variable Data Item Mapping

Variable Name	*Class*	*Format*
CtrlJobID	DVVAL	<OBJID>
QueueAvailableSpace	SV	5() (U1, U2, U4, U8)
QueuedCJobs	SV	L,n n = number of queued control jobs 1. <CtrlJobID$_1$> … n. <CtrlJobID$_n$>
SetUpName	ECV	20 (A[1..80]) Zero length string is used by the equipment to indicate 'unknown'.

7 SECS-II Attribute Definitions

7.1 *ControlJob Object SECS-II Attributes Definitions*

7.1.1 The following are the SECS-II structure definitions for the SEMI E94 ControlJob Object.

Table 6 ControlJob Object Attribute Definitions

Attribute Name	*Attribute Data Form: SECS-II Structure*
ObjType	20 (A) 'ControlJob'
ObjID	<OBJID> CtrlJobID (Conforms to the restrictions of ObjID as specified in SEMI E39.1, § 6.)
CarrierInputSpec	L,n n = number of input carriers 1. <CARRIERID$_1$> … n. <CARRIERID$_n$>
CurrentPRJob	L,n n = number of process jobs 1. <PRJOBID$_1$> … n. <PRJOBID$_n$>
DataCollectionPlan	20 (A) DataCollectionPlan

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Attribute Name	*Attribute Data Form: SECS-II Structure*
MtrlOutByStatus	L,n n = number of status dispositions 1. L,2 1. 51 (U1) $MaterialStatus_1$ 2. L,2 $DestinationMap_1$ 1. <$CARRIERID_{1,1}$> 2. L,m m = number of slots 1. <$SLOTID_1$> … m. <$SLOTID_m$> … n. L,2 1. 51 (U1) $MaterialStatus_n$ 2. L,2 $DestinationMap_n$ 1. <$CARRIERID_{n,1}$> 2. L,m m = number of slots 1. <$SLOTID_1$> … m. <$SLOTID_m$> If m = 0, the substrates may be placed by the equipment in any available slot.
MtrlOutSpec	L,p p = number of mapping source/destination pairs*) 1. L,2 1. L,2 $SourceMap_1$ 1. <$CARRIERID_{1,1}$> 2. L,m m = number of slots 1. <$SLOTID_1$> … m. <$SLOTID_m$> 2. L,2 $DestinationMap_1$**) 1. <$CARRIERID_{1,2}$> 2. L,m m = number of slots 1. <$SLOTID_1$> … m. <$SLOTID_m$> … p. L,2 1. L,2 $SourceMap_p$ 1. <$CARRIERID_{p,1}$> 2. L,n n = number of slots 1. <$SLOTID_1$> … n. <$SLOTID_n$> 2. L,2 $DestinationMap_p$ 1. <$CARRIERID_{p,2}$> 2. L,n n = number of slots 1. <$SLOTID_1$> … n. <$SLOTID_n$>

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Attribute Name	*Attribute Data Form: SECS-II Structure*
MtrlOutSpec cont.	* If the number p of source/destination pairs is zero, material shall be returned to its source location in its source carrier after processing. The only exception to this requirement is in the case of wafer sorters, where jobs such as Randomization are permitted to change the destination locations even though MtrlOutSpec is empty. ** To implement the optional support for a DestinationMap that is unknown at job creation, a DestinationMap contained in a MtrlOutSpec is allowed to be an empty structure, in which case its SECS-II structure shall be: L,0 DestinationMap
PauseEvent	L,n n = number of collection events 1. <CEID$_1$> … n. <CEID$_n$>
ProcessingCtrlSpec	L,p p = number of process jobs assigned to control job 1. L,3 1. <PRJOBID$_1$> 2. L,m ControlRule$_1$ 1. L,2 1. 20 (A) RuleName$_1$ 2. (Any format) RuleValue$_1$ … m. L,2 1. 20 (A) RuleName$_m$ 2. (Any format) RuleValue$_m$ 3. L,n OutputRule$_1$ 1. L,2 1. 51 (U1) MaterialStatus$_1$ 2. (Any format) OutputRuleValue$_1$ … n. L,2 1. 51 (U1) MaterialStatus$_n$ 2. (Any format) OutputRuleValue$_n$ … p. L,3 1. <PRJOBID$_p$> 2. L,m ControlRule$_p$ 1. L,2 1. 20 (A) RuleName$_1$ 2. (Any format) RuleValue$_1$ … m. L,2 1. 20 (A) RuleName$_m$ 2. (Any format) RuleValue$_m$ 3. L,n OutputRule$_p$ 1. L,2 1. 51 (U1) MaterialStatus$_1$ 2. (Any format) OutputRuleValue$_1$ … n. L,2 1. 51 (U1) MaterialStatus$_n$ 2. (Any format) OutputRuleValue$_n$

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Attribute Name	*Attribute Data Form: SECS-II Structure*
ProcessingCtrlSpec cont.	ControlRule 1) m = 0 indicates no ControlRule specified. 2) RuleName and RuleValue are equipment specific. OutputRule 1) n = 0 indicates no OutputRule specified. 2) There is no defined OutputRuleValue. The content and format of OutputRuleValue is equipment dependent per SEMI E94 intent.
ProcessOrderMgmt	51 (U1) ProcessOrderMgmt ProcessOrderMgmt is enumerated as follows: 1 = ARRIVAL 2 = OPTIMIZE 3 = LIST
PRJobStatusList	L,n n = number of process jobs defined in the control job L,2 <$PRJOBID_1$> <$PRSTATE_1$ >: L,2 <$PRJOBID_n$> <$PRSTATE_n$>
StartMethod	11 (BOOLEAN) StartMethod StartMethod is as follows: TRUE – Auto FALSE – UserStart
State	51 (U1) State State is enumerated as follows: 0 = QUEUED 1 = SELECTED 2 = WAITINGFORSTART 3 = EXECUTING 4 = PAUSED 5 = COMPLETED

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